ML Flow → Basic EDA → Identify Feature/Target → Data Splitting → Model Training → Model Testing → Optimization & Deploy

Data Preparation for ML Model.

© Polynomial Regression (one or more feature) [Degree]

$$Y = m_1 x + m_2 x^2 + m_3 x^3 + \cdots + c$$

⇒ from sklearn.preprocessing import PolynomialFeatures

ex: a, b

a, b, a^2, b^2, a.b

X_{train}, X_{test}

Step 1: `poly = PolynomialFeatures(degree=2)` → create object with degree

S2: `X_train_poly = poly.fit_transform(X_train)` → Convert training features

S3: `X_test_poly = poly.transform(X_test)` → Convert testing features too

```
from sklearn.linearmodel import LinearRegression
model = LinearRegression()                          → Create model object
model.fit(X_train_poly, Y_train)                    → Model training using poly features
```

Encoding

1. LabelEncoder → Categories are in an order
2. OneHotEncoder → Categories are nominal
3. MinMaxScaler → Conversion with Scaling (Standardized Conversion)

```
from sklearn.preprocessing import LabelEncoder, OneHotEncoder,
                                  MinMaxScaler
```

Problem with: incorrect selection / insufficient data / insufficient features / incorrect sampling / incorrect preprocessing.

① Overfitting → not good with new data (model accuracy: High)
(training)

② Underfitting → not good with testing data & new data.
(training) (mode accuracy: Low)

Note: model accuracy does not guarantee that model is perfect. We might have overfitting issue.